

Nishant Kumar

Computational Biologist - AI for Protein Function & Immunology
nishantk@iiitd.ac.in - +91-8755198690 - GitHub - Google Scholar - Website

Research Summary

Computational biologist developing machine-learning methods to understand and engineer protein and peptide function in immune and disease contexts. My research focuses on predictive modeling of antigenicity, antibody–protein interactions, protein stability, and therapeutic peptides using protein language model embeddings (ESM/ProtBERT) combined with interpretable machine learning. I build curated biological datasets, benchmark models, and deploy reproducible tools (web servers and packages) used by experimental researchers for hypothesis generation and therapeutic design.

Education

Ph.D. in Computational Biology	8.71/10 CGPA
Indraprastha Institute of Information Technology Delhi (IIIT-Delhi), India	2021 – Present
M.Sc. in Biotechnology	
Ch. Charan Singh University, Meerut, India	77.20% (2019)
B.Sc. in Biotechnology	
Ch. Charan Singh University, Meerut, India	68.80% (2017)

Publications

Peer-reviewed journal

- **HAIRPred: Prediction of antibody-interacting residues**
R. Sahni, **Nishant Kumar**, Raghava GPS Protein Science, 2025
- **Prediction of anti-freezing proteins from their evolutionary profile**
Kumar N, Choudhury S, Bajiya N, Patiyal S, Raghava GPS Proteomics, 2024
- **CLBTope: Predicting of linear and conformational B-cell epitopes from sequence**
Kumar N, Tripathi S, Sharma N, Patiyal S, Devi NL, Raghava GPS Computers in Biology and Medicine, 2024
- **Multi-perspectives and challenges in identifying B-cell epitopes**
Kumar N, Bajiya N, Patiyal S, Raghava GPS Protein Science, 2023
- **DMPPred: identification of antigenic regions responsible for inducing T1 diabetes mellitus**
Kumar N, Patiyal S, Choudhury S, Tomer R, Dhall A, Raghava GPS Briefings in Bioinformatics, 2023
- **Prediction and design of thermostable proteins with a desired melting temperature**
Tijare P, **Kumar N**, Raghava GPS (*Joint First Author) Scientific Reports, 2025
- **Prediction of inhibitory peptides against *E. coli* with desired MIC value**
Bajiya N, **Kumar N**, Raghava GPS Scientific Reports, 2025
- **Prediction of hemolytic peptides and their hemolytic concentration**
Rathore AS, **Kumar N**, Choudhury S, Mehta NK, Raghava GPS Communications Biology, 2025
- **PlantDRPpred: Prediction of plant resistance proteins**
Gahlot PS, Choudhury S, Bajiya N, **Kumar N**, Raghava GPS Proteomics, 2025
- **HLA-DR4Pred2: An improved method for predicting HLA-DRB1*04:01 binders**
Patiyal S, Dhall A, **Kumar N**, Raghava GPS Methods, 2024
- **Dual blood-brain barrier-glioma targeting peptide-poly(levodopamine) hybrid nanoplat-forms as potential near infrared phototheranostic agents in glioblastoma**
Dube T, **Kumar N**, Bishnoi M, Panda JJ Bioconjugate Chemistry, 2021

Preprints

- **MAP format for representing chemical modifications, annotations, and mutations in protein sequences: An extension of the FASTA format**
Shendre A, Mehta NK, Rathore AS, **Kumar N**, Patiyal S, Raghava GPS arXiv (q-bio.BM), 2025
- **In silico tool for identification of colorectal cancer from cell-free DNA biomarkers**
Mathur K, Jain S, Bajiya N, **Kumar N**, Raghava GPS arXiv (q-bio.GN), 2025
- **PEPlife2: An updated repository of the half-life of peptides**
Alam U, Chaudhary K, **Kumar N**, Tomer R, Patiyal S, Raghava GPS bioRxiv, 2025
- **CBTOPE2: An improved method for predicting of conformational B-cell epitopes**
Pandey A, Megha, **Kumar N**, Sahni R, Raghava GPS arXiv (q-bio.BM), 2025

Research Experience

July 2021 -IIIT-Delhi, India

current *PhD Thesis:- Development of AI-based tools for designing subunit vaccines*

Advisor: Prof. Gajendra P.S. Raghava

Research: Curated and benchmarked experimental datasets; built deployable ML/DL pipelines; and applied state-of-the-art techniques (ensemble learning, protein language-model embeddings: ProtBERT/ESM) for tasks including B-cell epitope prediction, antifreeze proteins, protein thermostability, and Type 1 diabetes-related immunoinformatics.

Dec.2020 - Premas Biotech Pvt. Ltd, Manesar, Gurugram (Haryana)

June 2021 *Project:- “AXTEX-4D” in Mammalian Cell Culture Lab Department*

Supervisor: Dr. Ambica Baru

Responsibilities: Cell line maintenance, data analysis, 3D culture, Spheroids formation by using hanging drop method.

Jan.2019 - Institute Of Nanoscience & Technology, Mohali (Punjab)

July 2019 *Project:- Peptide-Based Theranostics In Treating Glioblastomas*

Supervisor: Dr. Jiban Jyoti Panda (Scientist ‘E’)

Responsibilities: Setting up instruments and laboratory equipment used to conduct and monitor experiments such as Centrifuges, Spectrophotometer, and Zeta sizer Nano ZSP.

Technical Expertise

Programming & Computing: Python, Bash/Shell, Linux/Unix, HPC environments, R, PHP, SQL

Data Analysis & Scientific Libraries: NumPy, Pandas, scikit-learn, SciPy, Matplotlib, TensorFlow, PyTorch, Jupyter Notebook

Bioinformatics & Omics Analysis: Sequence analysis, dataset curation, protein feature extraction, LLM (ProtBERT, ESM2), clustering, dimensionality reduction, biological signature discovery

NGS & Multi-Omics RNA-seq preprocessing concepts, quality control, normalization, differential analysis concepts, single-cell analysis workflows (QC → normalization → clustering → annotation)

Reproducible Research Pipelines Docker, Singularity, version-controlled pipelines, workflows

Biological Databases GEO, UniProt/Swiss-Prot, PDB, IEDB, Metlome, PANC-DB

Web & Software Deployment Git/GitHub, PyPI package development, standalone bioinformatics tools, web servers (MySQL, HTML/CSS/JavaScript),

Wet-Lab Familiarity Mammalian cell culture, cell line maintenance, 3D spheroid culture, drug treatment assays, FACS exposure, spectrophotometry, microscopy

Instruments UV-Vis spectrophotometer, Zetasizer Nano ZSP, Bright Field Microscope, FACS (Fluorescence-activated cell sorting), Centrifuge Machine, Laminar Air Flow

Mentoring & Teaching Experience

Teaching Assistant, IIIT-Delhi, India

2022-2023

- **Courses:** Machine Learning in Biomedical Applications, and Big Data Mining in Healthcare
- Conducted tutorials and problem-solving on ML methods, evaluation, and biological data analysis
- Guided students in implementing ML pipelines and interpreting biological datasets
- Graded assignments and provided structured feedback on computational analyses

Research Mentorship

- Mentored 4+ students in thesis projects involving bioinformatics, machine learning, and data analysis
- Assisted students with dataset preprocessing, model development, validation, and scientific writing
- Supervised project design, troubleshooting, and result interpretation, and developemnt of workflows

Conference and Achievements

1. **Presented - DMPPRed at JBI2025 XV Bioinformatics Symposium (JBI2025)** at Madrid, Spain, 22 – 24 October 2025
2. **Presented - Multi-perspectives and challenges in identifying B-cell epitopes** at IIIT-Delhi (RIISE), 15-16 March, 2024
3. **Adipo-Cible Winter School — Obesity Unraveled: From Cellular Mechanisms to Therapeutic Pathways** held by Université Côte d’Azur, France, Dec. 16–19, 2024.
4. **InBix–ADNAT 2024 Conference:** Integrating Archaeogenetics, Forensics & Multi-omics for Human Health Breakthroughs, 28–30 November 2024, Banaras Hindu University, Varanasi, India.
5. **3-day Hands-on Workshop:** “Introductory Omics and RNA-seq Data Analysis,” 11–13 July 2024, organised by Bio-IT Centre, IBAB, Bengaluru, India.
6. **Workshop:** Establishing and Sustaining Cloud Infrastructure for Informatics-based Healthcare, 22–23 March 2024, IIIT-Delhi, India.
7. **CSIR-UGC JRF Fellowship (2020)**

Competitions

1. **DREAM Challenge:** FINRISK - Predicting heart failure using gut microbiome and clinical data.
2. **Kaggle Competition:** Novozymes enzyme stability prediction, thermostable mutations in enzymes

Referees

Name Prof. Gajendra P.S. Raghava
Contact raghava@iiitd.ac.in
Tel +91-11-26907444
Address IIIT-Delhi, India
Position Director, IIIT-Delhi
Web Web Profile
Scholar Google Scholar

Name Prof. Ganesh Bagler
Contact bagler@iiitd.ac.in
Tel +91-11-26907443
Address IIIT-Delhi, India
Position Head & Professor (CB)
Web Web Profile

Name Dr. Gaurav Ahuja
Contact gaurav.ahuja@iiitd.ac.in
Tel +91-11-26907475
Address IIIT-Delhi, India
Position Associate Professor (CB)
Web Web Profile

Name Dr. Ambica Baru
Contact ambica.baru@premasbiotech.com
Address Premas Biotech Pvt. Ltd., Haryana, India
Position Scientist (Mammalian Cell Culture Lab)